Metagenomic Analysis of Microbial Genera on Solar Panels

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Background

Solar panels provide a renewable energy source that reduces the dependency on fossil fuels.[1] However, their energy output can be hindered by the accumulation of environmental particles, which affects the light absorption of solar panels.[2] Environmental particles, such as soil and pollen, were believed to be the primary contributors to this loss of efficiency. However, after imaging solar panels (Figure 1), it is hypothesized that microbial growth also affect solar panel efficiency.[1,3]

To address this, I am collaborating with Solar Unoiled[4] to conduct a metagenomic analysis of solar panels in North Carolina, using a PromethION sequencer and several long-read pipelines to determine the microbial composition.

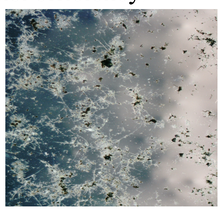


Figure 1: Image of solar panel

Objectives

1. Characterize the fungal genera present on solar panels
2. Investigate genera characteristics to understand what traits lead to survival on solar panels

Methods

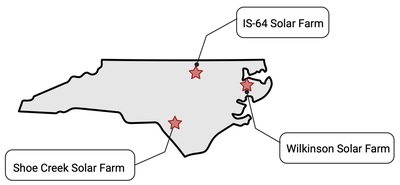


Figure 2: Sites locations in NC



Figure 3: Swabbing procedure

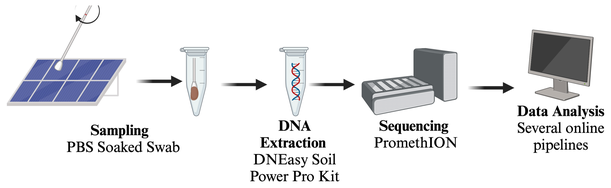


Figure 4: Schematic of data collection and processing

Site Locations

Site Name	Location	Nomenclature	# of Samples	# Unique Solar Panels
IS - 64	Hope Mills, NC	IS	6	2
Wilkinson	Wilkinson, NC	Wilk	8 (1 lost)	3
Shoe Creek	Laurinburg	SC	6/9	2/3

Test Run Site Location

Site Name	Location	Nomenclature	# of Samples	# Unique Solar Panels
Shoe Creek	Laurinburg	SC	3	1

Table 1: Table showing different sites and number of samples per site, showing the samples used as the test run from Shoe Creek

Results

The Dorado[5] and pb-metagenomics-tools[6] pipelines were used to take the PromethION raw reads and turn them into Megan reports, which were then used to make the graphs below.

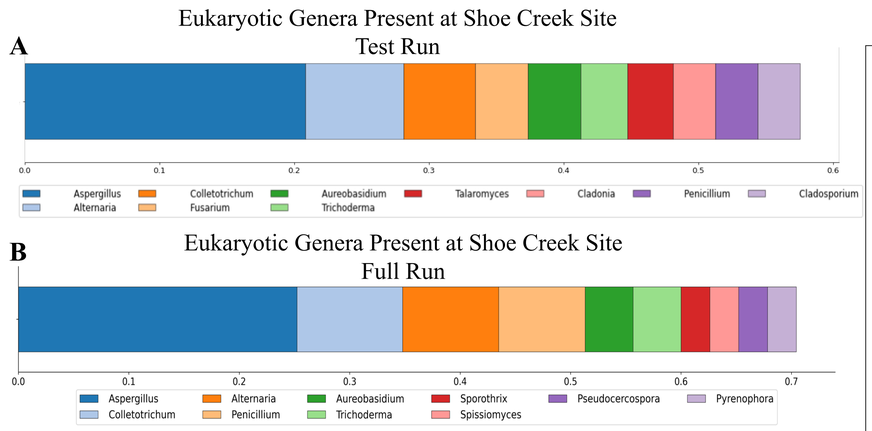


Figure 5 (A & B): This figure shows the metagenomic composition of fungi for the Shoe Creek test run (A) and the actual run (B). Samples were taken at the same time and split to see if the composition was preserved across runs to test the efficacy of the protocols.

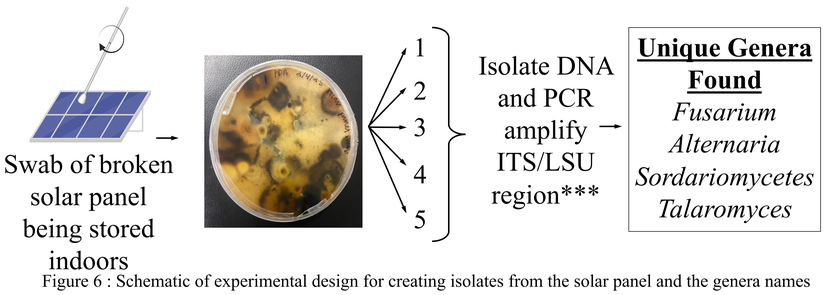


Figure 6 : Schematic of experimental design for creating isolates from the solar panel and the genera names of the isolates found

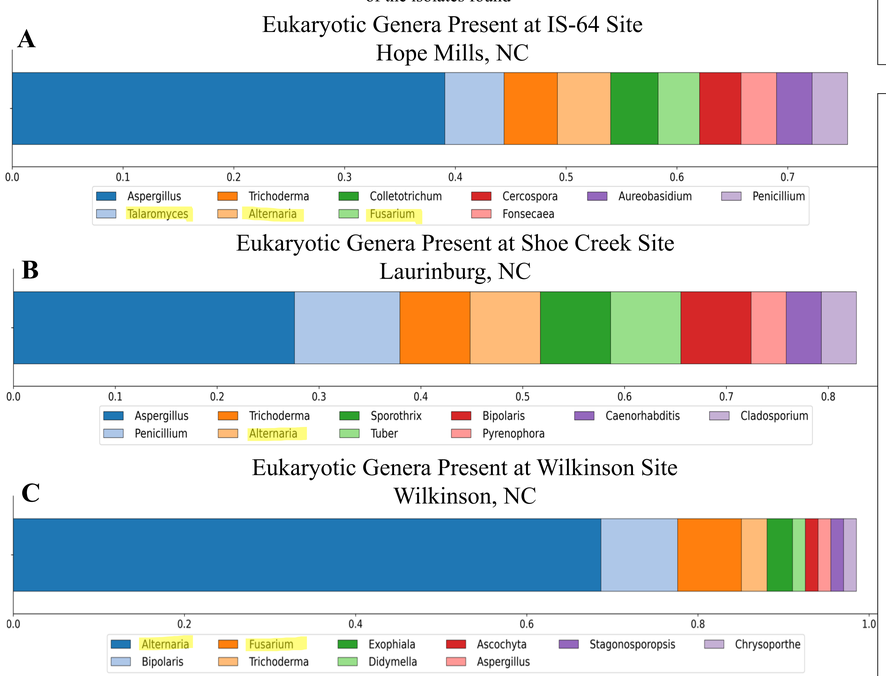


Figure 7 (A-C): Figure of the metagenomic composition of fungal genera at (A) IS-64, (B) Shoe Creek, and (C) Wilkinson. Highlighted genera indicate that the genus was isolated from the broken solar panel.

Results

Nine samples from Shoe Creek were collected on a single day. Three were used as a test run, and the remaining six were used in the actual run to compare results.

Microbial composition is relatively similar between the same site on different runs, indicating that the PromethION ran accurately shows the microbial community

A broken solar panel stored in a garage for 6 weeks was swabbed and cultured on PDA plates to see if the genera isolated match what was identified from the PromethION. *Fusarium*, *Alternaria*, and *Talaromyces* are seen in the PromethION results. Sordariomycetes is a class-level identification and is not shown in the results

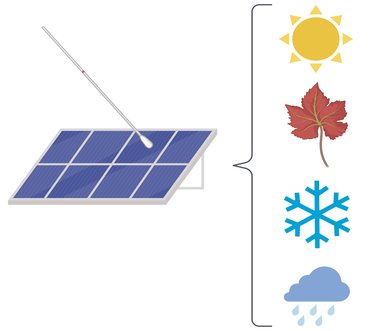
Results show some variability between sites but have mostly similar microbial compositions.

Common Genera Across All Sites

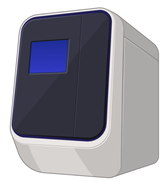
- *Aspergillus*
- *Alternaria*
- *Fusarium*
- *Trichoderma*

These genera are found to be UV, heat, radiation resistant, and exhibit increased melanin production.[1,3] These traits enable the fungi to survive, explaining why we observe growth in seemingly uninhabitable environments.

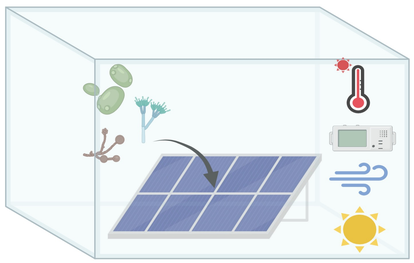
Future Directions



Look at microbial composition shifts based on season



Quantify the abundance of fungi versus bacteria using qPCR



Regrow solar panel isolates in an environment that simulates solar panel conditions

References and Acknowledgements

[1] Pocar et al., (2018) *Frontiers in Microbiology* [2] Al-Sharafi et al., (2024). *Cleaner Energy Systems* [3]Tanner et al., (2017). *Environmental Microbiology* [4] Solar Unsoiled (2025) [5] Oxford Nanopore Technologies. (n.d.). *Dorado: Oxford Nanopore's Basecaller*. GitHub.[6] Pacific Biosciences. (n.d.). *Taxonomic Profiling Tutorial: Minimap2 + MEGAN*. GitHub.

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